

SHIVAM MISHRA

Software Development Engineer (SDE) — Full-Stack, Backend & Applied ML

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PROFESSIONAL SUMMARY

Computer Science (AI) undergraduate with a strong foundation in data structures, algorithms, and system design, building scalable backend systems and ML-powered applications. Skilled in REST API design, authentication, relational databases, distributed real-time services, and deep learning with PyTorch. Focused on turning everyday problems into working, real-world software. Seeking an SDE internship to work on real-world software products.

EDUCATION

Manipal Institute of Technology — B.Tech, Computer Science (AI Specialization)	08/2024 – Present CGPA: 8.5 / 10
City Montessori School — Intermediate (Class 12)	04/2023 – 06/2024 93.5%
City Montessori School — Matriculation (Class 10)	04/2021 – 06/2022 96.6%

PERSONAL PROJECTS

StockSim — Full-Stack Virtual Stock Trading Platform [Live Demo](#) | [GitHub](#)

- Built a virtual NSE trading platform with INR 1,00,000 virtual capital using live prices from the Yahoo Finance API, with JWT authentication, weighted average buy-price calculation, and real-time portfolio profit and loss tracking.
 - Developed a leaderboard, watchlist, and transaction history feature using MVC architecture with Express.js.
- Technologies: Node.js, Express.js, MySQL, EJS, JWT, bcrypt, Yahoo Finance API, Render, Clever Cloud, CSS, JavaScript*

Nexus — Distributed Real-Time Multiplayer Game Server [Website](#) | [GitHub](#)

- Built a distributed, horizontally-scaled multiplayer game server from scratch with matchmaking decoupled from game hosting, coordinated across instances via Redis pub/sub and TTL-based heartbeat service discovery.
 - Implemented an authoritative 20Hz tick-based simulation with client-side prediction, server-side lag compensation using position-history rewinding, and stateful reconnection with grace-period token resumption.
 - Built a load-testing script simulating dozens of concurrent players and a live observability dashboard exposing per-process metrics (tick timing, queue depth, load).
- Technologies: Node.js, WebSockets (ws), Redis, Vanilla JavaScript, Canvas*

NutriVision — AI Food Recognition & Calorie Estimator [Live Demo](#) | [GitHub](#)

- Developed a deep-learning based food recognition application using MobileNetV3 and PyTorch, supporting 15 Indian food categories and 101 Food-101 global categories.
 - Built an interactive Streamlit web application for image-based food classification, confidence scoring, portion-based calorie estimation, and nutritional information visualization.
 - Implemented separate Indian and global food recognition pipelines, model evaluation scripts, preprocessing utilities, and integrated trained models for real-time inference.
 - Deployed the app on Streamlit Community Cloud and maintained a reproducible setup using GitHub, Python virtual environments, and requirements management.
- Technologies: Python, PyTorch, Torchvision, MobileNetV3, Streamlit, Plotly, Pandas, NumPy, Pillow*

SKILLS

Programming Languages: Java, Python, C, JavaScript, SQL

Frontend: HTML, CSS, JavaScript, EJS, Canvas

Backend & Systems: Node.js, Express.js, REST APIs, MVC, JWT, bcrypt, WebSockets, Redis (pub/sub)

Database: MySQL, DBMS

Artificial Intelligence & ML: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Neural Networks, Transfer Learning, PyTorch, Torchvision, Scikit-learn, NumPy, Pandas

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, System Design

Tools & Platforms: Git, GitHub, Postman, Streamlit, Plotly, Render, Clever Cloud

CERTIFICATIONS & ACHIEVEMENTS

- HackVega 2.0 — National Hackathon**, Certificate of Participation (Careernet/myCareernet); competed among 47,000+ engineering students from the 2026–2029 graduating cohorts.
- Solved 200+ Data Structures & Algorithms problems covering Arrays, Stacks, Trees, Binary Search Trees, and Heaps; weekly LeetCode contest participant.

LANGUAGES

English (Full Professional Proficiency) | Hindi (Full Professional Proficiency)